

PROJECT AEGIS

Sovereign Autonomous Systems for Infantry & Medical Support

Authors: Robert Charest & Andrew Ross | April 12, 2026

Executive Summary

Project AEGIS delivers a sovereign autonomous system for the Canadian Armed Forces, centered on the LTPE v6.3 navigation kernel. The core innovation is the **Ritual Layer**, implemented as an **Asynchronous Deterministic Fallback (ADF)**. This mechanism enables the system to maintain reliable operation by rapidly resolving logic entropy and reverting to pre-validated high-confidence behaviors when sensor fusion degrades under electronic warfare, jamming, or environmental interference.

Validation was conducted using Monte Carlo Simulation of Simulation (SoS) with 2,000 trials per scenario, modeling interacting air and ground swarms with low-bandwidth LoS Hub data seeding. Results demonstrate exceptional resilience and graceful degradation:

Condition	Success Rate (Baseline)	Success Rate (LTPE v6.3)	Avg Global Stalls (LTPE)
Contested (40% dropout)	46.7%	97.0%	0.03
Catastrophic (90% dropout)	38.7%	94.4%	0.06

The LTPE v6.3 architecture demonstrates strong **graceful degradation**, maintaining high mission success and near-zero stall rates even under severe sensor degradation and full electromagnetic control. This capability directly addresses a critical gap for the Canadian Armed Forces by transforming autonomous platforms from potential liabilities into dependable, attritable assets suitable for Arctic and forward contested operations.

Full technical details, LTPE kernel source code, and Monte Carlo SoS validation scripts are available at:

github.com/Glargod/LTPE